# Magnetite-Free Sn-Doped Hematite Nanoflake Layers for Enhanced Photoelectrochemical Water Splitting

Hyo-Jin Ahn,[a, b, c] Stepan Kment,*[a, d] Jeong Eun Yoo,[b] Nhat Truong Nguyen,[b] Alberto Naldoni,[a] Radek Zboril,[a, d] and Patrik Schmuki*[a, b]

In the present work, we report a preparation strategy for hematite phase-pure photoanodes consisting of Sn-doped hematite nanoflakes/hematite thin film bilayer nanostructure (Sn-HB). This approach is based on a two-step annealing process of pure iron films deposited on fluorine doped tin oxide (FTO) substrates by advanced magnetron sputtering. While the high density hematite ultrathin nanoflakes (HNs) with detrimental iron oxide layers ($Fe_3O_4$ and/or FeO) are generated during the first annealing step at 400 °C for two hours, the second thermal treatment at 800 °C for 15 minutes oxidises all the undesired iron oxide phases to a photoactive hematite layer as well as is providing efficient Sn doping of a drop-casted $SnCl_4$ in order to increase the conductivity. The optimized Sn-HB shows an around 11 times higher photocurrent density (0.71 mA cm$^{-2}$ at 1.23 $V_{RHE}$) compared with a reference hematite photoanode produced from iron foil under the same conditions.

## Introduction

Due to environmental concerns regarding the use of fossil fuels, over the past decades renewable energy resources have received great attention.[1] Among them, photoelectrochemical (PEC) water splitting is a promising green approach as it leads to oxygen and hydrogen gases solely by using an aqueous electrolyte and solar energy.[2,3] A variety of oxide semiconductors, such as $TiO_2$,[4] $\alpha$-$Fe_2O_3$,[5] ZnO,[6] $Ta_3N_5$,[7] $WO_3$,[8] and $BiVO_4$,[9] have been used as photoanodes. In particular, $\alpha$-$Fe_2O_3$ (i.e., hematite) is one of the most promising photoanode materials because of a band gap (2.1 eV), which allows absorption of a significant portion of visible solar light and thus the high theoretical solar-to-hydrogen (STH) efficiency of 15~16%. The material has a high (photo)chemical stability, can be fabricated by simple processing, and bears low cost.[10–12] Nevertheless, the real PEC performance of hematite is still limited owing to a high recombination rate of photocharges resulting from a short photogenerated hole diffusion length ($L_h \approx 2$–4 nm), slow oxygen evolution reaction (OER) kinetics, and often a limited charge transfer at internal interfaces of photoanodes.[13–15]

While addressing these drawbacks, a large number of approaches including the fabrication of nanostructured hematite,[16] heteroatom-doping,[17] and surface catalytic modifications[18] have been reported to enhance the efficiency of photoelectrodes.[19] The beneficial effects of each strategy is roughly summarized in the following part. First, one dimensional (1D) nanoscale hematite structures (e.g., nanorods, nanotubes, nanoflakes)[20–23] increase the surface area for water oxidation and reduce the recombination of photoproduced charges within 1D structure because of an orthogonal charge separation with a strongly decreased diffusion path for photoexcited holes. Second, suitable elemental doping (e.g., $Ti^{4+}$,[24] $Sn^{4+}$,[20] $Si^{4+}$,[25] and $Pt^{4+}$[12]) increases the donor density of hematite, enhances the electrical conductivity, and can therefore improve the PEC performance of a photoanode. Third, surface modifications using co-catalysts (e.g., Co–Pi,[26] IrO,[11] FeOOH[21] and layered double hydroxides[14]) improve the OER kinetics of hematite, leading to a cathodic shift of the OER onset potential. The combination of these three key features significantly enhances the PEC performance of a hematite photoanode.[10,14,27]

Among $\alpha$-$Fe_2O_3$ nanostructures, hematite nanoflakes (HNs) show an ultra-thin and high-aspect-ratio morphology that has been considered an ideal nanostructure for PEC water splitting because of an anisotropic charge transfer of electron-hole-pairs (EHPs) in the hematite lattice.[28] However, HNs are synthesized from a metal substrate (such as a sheet or a foil) by thermal oxidation. This thermal oxidation step leads not only to hematite but also to magnetite ($Fe_3O_4$). This sub-oxide layer is inherent to high temperature oxidation of iron and forms underneath the hematite nanostructure at the interface to the

[a] Dr. H.-J. Ahn, Dr. S. Kment, Dr. A. Naldoni, Prof. R. Zboril, Prof. P. Schmuki
Regional Centre of Advanced Technologies and Materials and Czech Advanced Technology and Research Institute
Palacký University Olomouc
Šlechtitelů 27, Olomouc, 783 71, Czech Republic
E-mail: stepan.kment@upol.cz

[b] Dr. H.-J. Ahn, Dr. J. E. Yoo, Prof. N. Truong Nguyen, Prof. P. Schmuki
Department of Materials Science and Engineering
University of Erlangen-Nuremberg
Martensstrasse 7, D-91058 Erlangen, Germany
E-mail: Patrik.Schmuki@ww.uni-erlangen.de

[c] Dr. H.-J. Ahn
xEnergy and Catalyst
LSTME Busan Branch
31, Gwahaksandan 1-ro 60beon-gil, Gangseo-gu, 46742 Busan, Republic of Korea

[d] Dr. S. Kment, Prof. R. Zboril
Nanotechnology Centre, Centre of Energy and Environmental Technologies
VŠB – Technical University of Ostrava
708 00 Ostrava-Poruba, Czech Republic

Supporting information for this article is available on the WWW under https://doi.org/10.1002/celc.202200066

© 2022 The Authors. ChemElectroChem published by Wiley-VCH GmbH. This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.



© 2022 The Authors. ChemElectroChem published by Wiley-VCH GmbH.

metal.[29] This unavoidable magnetite layer strongly hampers the charge transfer from the hematite layer to the charge collector, thus representing one of the most important problems for the HNs photoanodes.[30,31] To overcome the drawbacks of such sub-oxide (magnetite) formation, several strategies have been reported, including Sn-ion doping of magnetite or attempts to add a gold layer between the magnetite layer and the iron foil.[28,32,33] However, these efforts cannot fully overcome the detrimental effects of the magnetite layer.

Herein, we introduce a novel two-step approach for the fabrication of magnetite-free hematite nanoflakes consisting of a bilayer structure of Sn-doped hematite nanoflakes/thin film (Sn-HB). The idea is to achieve nanoflake growth from an optimized finite layer of Fe, i.e., to prevent magnetite formation by achieving total oxidation of a thin Fe film, thus avoiding magnetite formation at the otherwise moving oxidation boundary. For this, we used thin Fe films (200–800 nm) deposited on fluorine doped tin oxide (FTO) glass by magnetron sputtering. Then the Fe thin film/FTO samples were placed in a furnace at 400 °C for synthesizing a nanoflake structure. In this intermediate state, we decorated the structures with Sn. A second thermal step at 800 °C was then applied for the drive-in of the doping and full oxidation of the nanoflakes. During this high-temperature annealing process, not only can $Sn^{4+}$ ions be incorporated in hematite lattice but also the remnants of sub-oxides are fully converted to $\alpha$-$Fe_2O_3$ because of the accelerated $O^{2-}$ diffusion at an elevated temperature and the absence of a metallic substrate that could form $Fe_3O_4$.[34] The optimized Sn-HB photoanode exhibited a photocurrent density of 0.71 mA cm$^{-2}$ at 1.23 $V_{RHE}$ (reversible hydrogen electrode, RHE); this represents a roughly 245% enhanced PEC activity over a conventional Sn-doped hematite structure on iron foil due to the absence of the detrimental $Fe_3O_4$ layers.

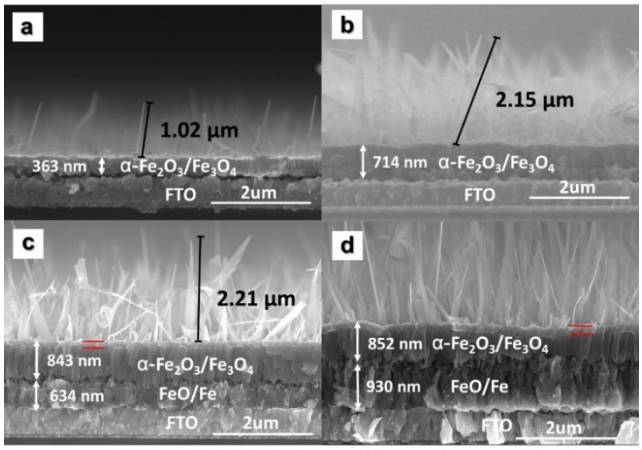



**Figure 1.** Cross sectional SEM images of (a) HNF-200, (b) HNF-400, (c) HNF-600, and (d) HNF-800, which are thermally oxidized 200, 400, 600, and 800 nm Fe films on the FTO glass at 400 °C for 2 h, respectively.

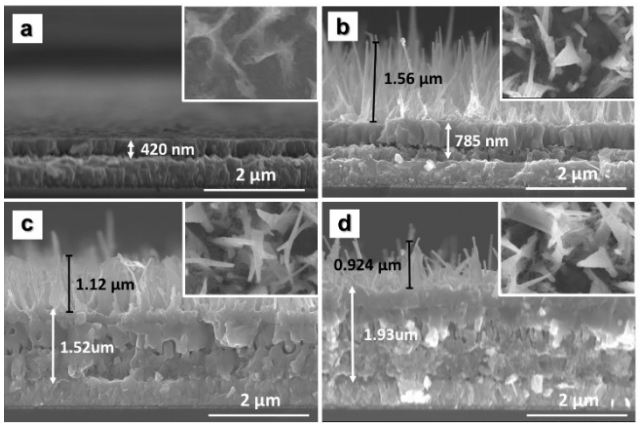



**Figure 2.** Cross sectional SEM images of (a) Sn-HB 200, (b) Sn-HB 400, (c) Sn-HB 600, and (d) Sn-HB 800, respectively. The insets in (a–d) are top view SEM images.

## Results and Discussion

To investigate the effect of the bottom layers ($Fe_3O_4$ and FeO) on the $\alpha$-$Fe_2O_3$ photoanode, metallic Fe layers with the thickness of 200, 400, 600, and 800 nm were deposited on the FTO glass by magnetron sputtering. The cross-section SEM images of the sputtered iron films are shown in Figure S1. The morphology of the deposited iron films was the same for all the samples since all the magnetron sputtering conditions were kept constant and only the time of the deposition varied. In a set of preliminary experiments and in accord with literature,[36] the optimal thermal growth condition for hematite nanoflakes is at 400 °C in air. During the thermal oxidation of the Fe thin film on the FTO glass in the furnace, the hematite nanoflakes (HNF) and the $Fe_3O_4$ layer grow as a result of the Fe and oxygen diffusion along the Fe grain boundaries.[35]

Figure 1 and Figure 2 shows the cross-sectional scanning electron microscopy (SEM) images of the HNF samples and Sn-HB samples fabricated from different thickness of the Fe thin films, respectively. The description of the sample designation codes is provided in the experimental part. The HNF-200 sample consists of a very low density of hematite nanoflakes (HNs) and 363 nm of a distinct layer comprising of $\alpha$-$Fe_2O_3$/$Fe_3O_4$ mixed-phase layer (Figure 1 and Figure S2), which has been identically reported in a number of studies.[28,29,33] The thickness of the $\alpha$-$Fe_2O_3$/$Fe_3O_4$ underneath the layer for HNF-400, HNF-600, HNF-800 samples (Figures 1b–1d) and for the HNF-iron foil (Figure S3) increased to 714, 843, 852, and 900 nm, respectively. These samples also showed much higher density of the HNs on their surfaces. On the other hand, the length and the morphology of the hematite nanoflakes were similar among the HNF samples regardless of the increase in the $\alpha$-$Fe_2O_3$/$Fe_3O_4$ bottom layer, as it can be seen in Figures 1b–1d and Figures S2–S3. Although a clear boundary between the $\alpha$-$Fe_2O_3$/$Fe_3O_4$ layer under the HNs can be observed, when the thickness of the Fe film is higher than 600 nm (red lines, Figure 1c–1d), there is no clear boundary between the $\alpha$-$Fe_2O_3$/$Fe_3O_4$ layers in the cross-sectional SEM images of HNF-200 (Figure 1a). However, as shown later in Figure 3, strong X-ray diffraction (XRD) peaks corresponding to magnetite phase can be observed in all HNF samples. Therefore, it is rational to define this bottom layer


© 2022 The Authors. ChemElectroChem published by Wiley-VCH GmbH.

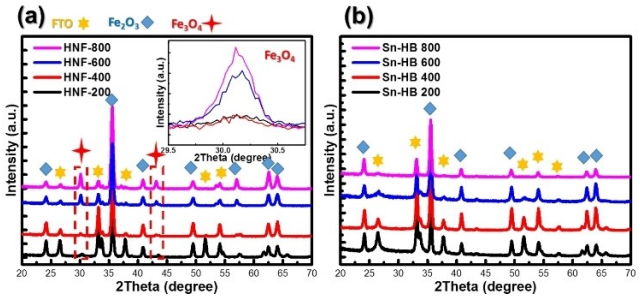



**Figure 3.** XRD patterns of (a) HNF-200, HNF-400, HNF-600, and HNF-800, respectively, and (b) Sn-HB 200, Sn-HB 400, Sn-HB 600, and Sn-HB 800, respectively. The inset in (a) is a magnified XRD peak for the magnetite (220) plane.

as a mixture of α-$Fe_2O_3$ and $Fe_3O_4$. It is noteworthy that the thickness of the α-$Fe_2O_3$/$Fe_3O_4$ layer does not change significantly when the thickness of the Fe film is higher than 600 nm. This is a consequence of the limited annealing time at 400 °C and the diffusion-limited oxide growth (see Figures 1c–d). Furthermore, the volume expansion of the Fe-oxide layers, which are thicker than the starting Fe seed layer, is also evident from the SEM images of HNF-600 and HNF-800, respectively.

Next, the Sn-doped hematite bilayer structures, which consisted of hematite nanoflakes and Sn-HB thin layers, were prepared by 20 μl of $SnCl_4$ solution treatment on the HNF samples followed by drying at room temperature for 10 min and a sintering process at 800 °C for 15 min. Finally, the Sn-HB samples were dipped in 1 M KOH solution for 12 h in order to remove the excess $SnO_2$ layers, as reported elsewhere.[36]

Figure 2 displays SEM images of the Sn-HB samples. The original length of the nanoflakes of ca. 2 μm on the HNF samples decreased to 1.56, 1.12, and 0.924 μm, respectively, after the annealing process. Moreover, the shape of the nanoflakes on the Sn-HB samples is also contracted compared to HNFs, which results from a structural deformation of hematite.[37,38] When HNF is converted to Sn-HB, the bottom layers (α-$Fe_2O_3$/$Fe_3O_4$ and/or FeO/Fe) in the HNF samples are fully oxidized to α-$Fe_2O_3$ in consequence of the high temperature annealing, which is evident from the XRD analysis (see Figure 3b). In addition, the thickness of the bottom layers increases to 420 nm, 785 nm, 1.52 μm, and 1.93 μm, respectively, as also evident in Figure 2. Although there were severe volume expansions, the detachment of the films was not observed (See Figure 2 and Figure S4). The morphology changes of Sn-HB 600 sample from Fe film to $Fe_2O_3$ bilayer structure was summarized in Figure S5. Furthermore, we confirmed that there is reduced reflection of hematite bilayer structure (Sn-HB 600) compared with single film structure (Sn-HB 200) due to hybrid structures, consisting of hematite nanoflakes and film (See Figure S6).

In order to study the crystalline structure of the samples, XRD measurements were performed. Figure 3a shows XRD patterns of the HNF samples with the typical reflections at 24.13°, 35.61°, 40.83°, 49.42°, 57.51°, 62.40°, and 63.98° corresponding to (012), (110), (113), (024), (122), (214), and (300) plane of hematite, respectively. The XRD peaks at 30.07° and 43.05° were assigned to (220) and (400) plane reflections of magnetite ($Fe_3O_4$) in the HNF samples.[28] The peak intensity of magnetite displayed in the inset of Figure 3a is well consistent with the thickness of the α-$Fe_2O_3$/$Fe_3O_4$ layer, as described above (Figure 1). After the second annealing step, the XRD peaks for magnetite in the Sn-HB samples (Figure 3b) disappeared, verifying that the magnetite phase had been fully transformed to hematite after the high temperature annealing process of the HNF samples. Since the oxygen diffusion rate increases with an elevated temperature, the bottom layers (α-$Fe_2O_3$/$Fe_3O_4$ and/or FeO/Fe) of the HNF are easily converted to hematite layers, as evident from Figure 2.

With the aim to examine the surface composition of the HNF and the Sn-HB samples, X-ray photoelectron spectroscopy (XPS) was carried out, and the corresponding high-resolution Sn 3d and Fe 2p spectra are shown in Figure 4. The XPS analysis of the Sn-HB samples shows two major peaks at 487.3 eV (Sn $3d_{5/2}$) and 495.7 eV (Sn $3d_{3/2}$), which are well consistent with the reported XPS data for Sn-doped hematite, demonstrating substitutional doping of $Sn^{4+}$ ions in the hematite lattice (see Figure 4(a)).[39] Figure 4b shows XPS peaks at 711 eV (Fe $2p_{3/2}$) and 724.5 eV (Fe $2p_{1/2}$), which were detected in Sn-HB 600 and HNF-600, respectively. Since $Sn^{4+}$ ions can reduce the $Fe^{3+}$ to $Fe^{2+}$, the XPS peak intensity of $Fe^{2+}$ (716 eV) for Sn-HB 600 was higher than that of HNF-600, suggesting a higher concentration of $Fe^{2+}$ ions on the surface of Sn-HB 600 than HNF-600.[40][41] The molar ratio of Sn and Fe obtained from the XPS analysis demonstrates an atomic percentage (Sn/(Sn + Fe)) of 0.03 % and 27.54 % for the HNF-600 and Sn-HB 600 (Table S1).

Figure 5a displays photocurrent density-potential (*J-V*) curves recorded in 1 M KOH electrolyte and by using AM1.5 chopped light for samples HNF-200, HNF-400, HNF-600, HNF-800, as well as for the HNF-iron foil. The photocurrent density of the HNF-200 sample is 0.021 mA cm$^{-2}$ at 1.23 $V_{RHE}$, which is similar to previously reported hematite thin film photoanodes. Considering the short light absorption length and short diffusion length of photoholes ($L_h \approx 2$–4 nm), the photocurrent density of HNF-200 with the thickness of 363 nm of $Fe_2O_3$/$Fe_3O_4$ thin film (Figure 2a) is accordingly low. After the growth of the HNs on the $Fe_2O_3$/$Fe_3O_4$ thin film, the photocurrent density of HNF-400 reaches 0.1 mA cm$^{-2}$ at 1.23 $V_{RHE}$. However, for thicker layers, the photocurrent density again decreases to 0.0675 mA cm$^{-2}$ (HNF-600), 0.0513 mA cm$^{-2}$ (HNF-800) and

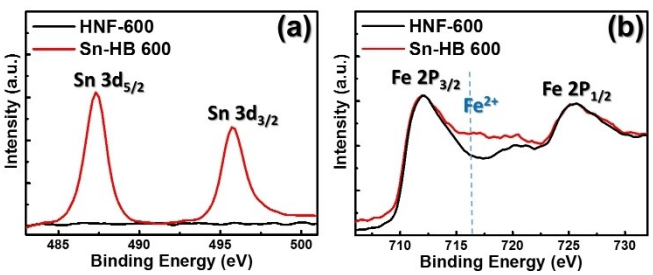



**Figure 4.** XPS spectra of (a) Sn 3d and (b) Fe 2p for HNF-600 and Sn-HB 600, respectively.


© 2022 The Authors. ChemElectroChem published by Wiley-VCH GmbH.

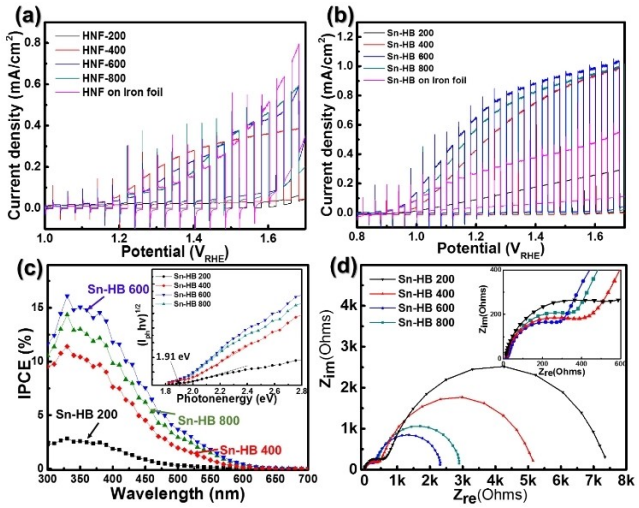



**Figure 5.** (a) *J-V* curve of HNF-200, HNF-400, HNF-600, HNF-800, and HNF on iron foil. (b) *J-V* curve of Sn-HB 200, Sn-HB 400, Sn-HB 600, Sn-HB 800, and Sn doped HNF on iron foil. (c) IPCE of Sn-HB 200, Sn-HB 400, Sn-HB 600, and Sn-HB 800 measured at 1.23 $V_{RHE}$. The inset shows the band gap calculation from a $(I_{ph}hv)^{1/2}$ vs. photonenergy ($hv$) plot. (d) Nyquist plots of Sn-HB 200, Sn-HB 400, Sn-HB 600, and Sn-HB 800 under 369 nm light-emitting diode (LED) source at 1.23 $V_{RHE}$. The Inset shows the enlarged high frequency region of Nyquist plots.

0.045 mA cm$^{-2}$ (HNF- iron foil) at 1.23 $V_{RHE}$. This trend is due to an increase in the thickness of the bottom layers ($Fe_2O_3$/$Fe_3O_4$ and FeO/Fe), as shown in Figure 2a and Figure 5a. Figure 5b further shows the *J-V* curves of the samples after the second thermal- and Sn-treatment step. An increase in the photocurrent density of HNF-600 to 0.71 mA cm$^{-2}$ at 1.23 $V_{RHE}$ was observed, after the high temperature annealing of the Sn-treated samples was carried out (Sn-HB 600). This drastic increase is due to the absence of the sub-oxide bottom layers and the enhanced conductivity introduced by the Sn doping (Figure 2b and Figure 5b). This can be confirmed by the Mott-Schottky plots which are well consistent with the PEC performances as shown in Figure S7. In the Mott-Schottky plots, the donor density of the Sn-HB 600 is higher than those of the other samples (see table in Figure S7) denoting higher donor concentration that might be another effect of the high density of the nanoflakes and thus efficient doping by Sn. For comparison, we attempted to achieve a similar high temperature Sn treatment using the samples without FTO. That is, the Sn-doped HNF on the iron foil sample was prepared by the corresponding 2-step annealing process (400 °C for 2 h and 800 °C for 5 min with the Sn treatment). The results are summarized in Figure S8. However, under these conditions, the volume expansion of the iron foil was too severe, and, therefore, the grown HNs finally partially peeled-off from the foil (see photograph in Figure S8d). When the thickness of the bottom hematite layer was higher than that of Sn-HB 600, the photocurrent density decreased to 0.63 mA cm$^{-2}$ (Sn-HB 800) and 0.29 mA cm$^{-2}$ (Sn-HB on iron film) at 1.23 $V_{RHE}$ because of a high recombination rate resulting from a longer electron pathway to the FTO glass. This can be observed in the *J-V* curve of Sn-HB 800 and Sn-HB on the iron foil (Figure 5b). The incident photon-to-current efficiency (IPCE) spectra in Figure 5c revealed a higher quantum efficiency of Sn-HB 600 compared to that of other samples at the applied potential of 1.23 $V_{RHE}$, which was well consistent with the J–V measurement results. The IPCE trend for all the samples followed the optical absorption of the HNs in the region of wavelengths between 300–600 nm. The maximum IPCE values of the Sn-HB samples can be found at the wavelength of 330 nm and it increased from 2.83 % (Sn-HB 200) to 16.10 % (Sn-HB 600), which was in line with the initial iron film thickness. Next, when the thickness of the initial iron film was higher than 600 nm, the maximum IPCE value decreased to 14.36 % (Sn-HB 800) because of a high recombination rate. The inset of Figure 5c shows the evaluation of the band-gap ($E_g$) from indirect electron transition ($(I_{ph}hv)^{1/2}$ vs. photon energy ($hv$)) of the Sn-HB samples leading to approximately 1.91 eV, which is well in the range of the reported $E_g$ of pristine hematite (1.9–2.1 eV).

To study the charge transfer kinetics of the hematite photoanodes, electrochemical impedance spectroscopy (EIS) was performed in 1 M KOH electrolyte at 1.23 $V_{RHE}$ applied potential, using a 369 nm light source (Figure 5d). The equivalent circuit model depicted in Figure S9 was used to fit the data of the Nyquist plots. This equivalent circuit represents trapping/de-trapping resistance of electrons in hematite, $R_1$; the space charge capacitance at the interface of the bulk hematite, $C_1$; the charge transfer resistance between the electrolyte and the surface of the hematite, $R_2$; the space charge capacitance at the interface between the electrolyte and the surface of the hematite, $C_2$; and the series resistance of the electrochemical cell, *Rs*.[17] The EIS fitting results of Sn-HB 200, Sn-HB 400, Sn-HB 600, and Sn-HB 800 are summarized in Table S1. The $R_s$ values for all samples are relatively low, which indicates an efficient transfer of electrons from the hematite and the FTO conductive electrode. The trapping/de-trapping resistance of electrons ($R_1$) and charge transfer resistance between the electrolyte/surface of the hematite ($R_2$) of the Sn-HB 200 defining the thin film structure show the highest values compared to the other Sn-HB samples, which means that the hematite thin film layer shows the poor PEC performance because of high recombination and poor charge extraction. It is interesting that even though Sn-HB 600 has a thicker bottom hematite layer compared with Sn-HB 200, the electron recombination rate in the bulk region ($R_1$) and the charge extraction resistance ($R_2$) of Sn-HB 600 is lower than that of Sn-HB 200, as displayed in Figure 5d. This may be attributed to the decreased recombination rate due to charge compensation of photo-generated holes in the hematite thin layer by additional electrons from HNs. When the thickness of the bottom layer reached 1.93 μm (Sn-HB 800), the $R_1$ and $R_2$ of Sn-HB 800 increased again owing to the prolonged pathway for electron transfer, which was consistent with the *J-V* curves of the Sn-HB samples.

From photoelectrochemical properties combined with impedance measurements of the Sn-HB samples, a charge transfer mechanism, as depicted in Figure 6, can be concluded. The generally high photocurrents of the hematite nanoflakes can be ascribed to the high anisotropic conductivity of hematite that is up to four orders of magnitude higher electron transport along


© 2022 The Authors. ChemElectroChem published by Wiley-VCH GmbH.

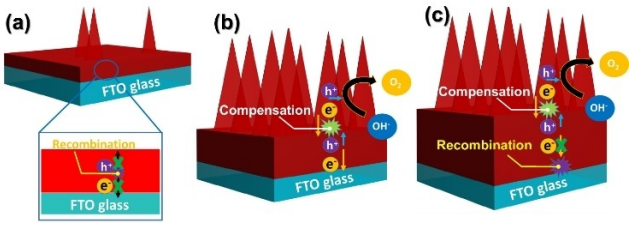



**Figure 6.** Schematic diagram for charge transfer mechanism of (a) Sn-HB 200, (b) Sn-HB 600 and (c) Sn-HB 800, respectively.

(110) hematite crystal planes than orthogonal to them and to the high aspect ratio structure of HNs.[42] In other words, not only can the photogenerated holes in the HNs easily reach the electrolyte through the thin nanoflake structure, but also the photogenerated electrons can flow to the FTO glass along the preferential (110) plane. We can thus suggest a possible charge transfer mechanisms considering also the thickness of the hematite bottom layer as well as the density of HNs grow, which can describe the enhanced PEC activity of the Sn-HB 600 sample. First, Figure 6a describes the charge transfer mechanism of the Sn-HB 200 sample. As shown above, this sample contains very low density of the HNs on the top of a relatively thick hematite layer. Because of generally very low electrical conductivity of hematite films and short diffusion length of holes, the EHPs generated in the bulk hematite layer easily recombine, which is similar to the typical hematite thin films. By contrast, when the density of the HNs considerably increases and the hematite bottom layer has the optimal thickness, as in the case of sample Sn-HB 600, the photogenerated holes in the bottom hematite thin film can be compensated by the photogenerated electrons from the upper HNs. Since the photoexcited holes from the HNs can be easily extracted to electrolyte, the electrons from the HNs can reach the bottom layer and act as a hole scavenger in the bulk hematite thin film. These charge compensations between the electrons from the HNs and the holes from the thin film layer can effectively reduce the probability of a recombination rate in the thin film layer and the HNs, as it is schematically depicted in Figure 6b. Finally, if the thickness of the bottom layer exceeds a certain limit and simultaneously the length of the HNs is shortened, the recombination rate increases again. It is a consequence of prolonging the electron pathway to the FTO charge collector and a poor compensation effect by a decreased number of electrons resulting from the reduced light absorption of the shortened HNs. Since the hole scavenger present in the electrolyte reduces the charge transfer resistance between hematite surface and electrolyte, our aforementioned mechanism can be confirmed by comparing the EIS data with and without the hole scavenger.[43] Figure S10a and S10b show the EIS data of the samples measured in 1 M KOH solution at 1.0 $V_{RHE}$ and under 1 sun illumination without and with 0.5 M $Na_2SO_3$ as the hole-scavenger, respectively. In both cases the sample Sn-HB 600 show the lowest resistance implying reduced recombination of charges also in the bulk hematite layer and thus the best charge transfer ability. In addition, the charge separation ($\eta_{sep}$) and charge injection ($\eta_{inj}$) efficiency of Sn-HB600 shows the highest value compared to that of other samples, indicating reduced charge recombination in photoanode (Figure S12). Similar mechanism can also be considered for the sample Sn-HB 800 as depicted in Figure 6c. This sample according to the SEM analysis (see Figure 2d) shows the thickness of the bottom hematite layer of 1.5 μm and the length of HNs of 1.12 μm. However, this sample provides slightly lower PEC activity presumably due to the detrimental effect of low absorption coefficient and short diffusion length of the charges, which again start to reduce the PEC performance.

As mentioned before, another limitation of hematite for an efficient application for PEC water splitting is poor oxygen evolution kinetics on its surface. Therefore, in order to further improve the PEC performance, an OER catalyst, zinc-cobalt layered double hydroxide (Zn–Co LDHs), was deposited on HNF-600 and Sn-HB 600 by immersing in the diluted Zn–Co LDH solution for 10 min as shown in Figure 7. The Zn–Co LDH co-catalyst was synthesized using the same method as reported elsewhere.[14] The SEM image (Figure S13) and XPS analysis (Figure S14) was used to confirm the presence and the proper chemical composition of the Zn–Co LDHs deposited on the surface of the Sn-HB 600 (Zn–Co LDH/Sn-HB 600), respectively. The high-resolution XPS peaks for Zn 2p and Co 2p of Zn–Co LDH in Figure S14 show the characteristics of Zn–Co LDH, which are in line with the literature.[14] The photocurrent of Zn–Co LDH/Sn-HB 600 in Figure 7 shows 0.97 mA $cm^{-2}$ at 1.23 $V_{RHE}$, which represents about 15 times enhanced PEC performance compared with HNF-600 (0.0675 mA $cm^{-2}$ at 1.23 $V_{RHE}$), i.e., the material can indeed be further improved by using an OER catalyst. Furthermore, the onset potential of Zn-CO LDH/Sn-HB 600 shows a 119 mV cathodic shift from 0.963 (Sn-HB 600) to 0.844 $V_{RHE}$. Overall, this considerable improvement is attributed to the facile charge transfer by effectively doped Sn ion, the absence of a poorly conductive $Fe_3O_4$ layer, and the enhanced OER property.

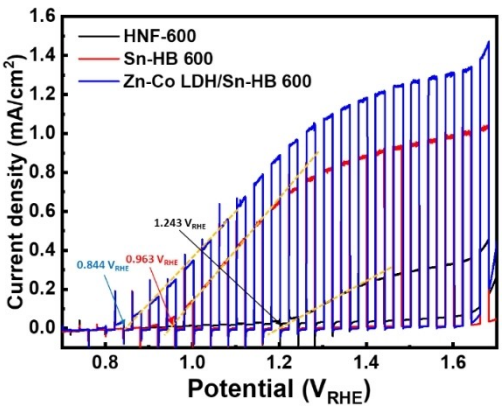



**Figure 7.** *J-V* curve of HNF-600, Sn-HB 600, Zn–Co LDH/Sn-HB 600, respectively.


© 2022 The Authors. ChemElectroChem published by Wiley-VCH GmbH.

## Conclusion

In conclusion, we fabricated an $Fe_3O_4$ layer free Sn-doped hematite bilayer structure of hematite nanoflakes/thin film by using first the magnetron sputtering of iron films with different thicknesses onto an FTO substrate followed by simple two-step annealing process. During the first step, the as-deposited Fe films were annealed at 400 °C for 2 h and then the Sn treatment of the hematite nanostructures was carried out, followed by second annealing at 800 °C for 15 min. Not only does our approach convert the high resistive bottom layers to a photo-responsive hematite layer, but it also makes a complementary relationship between the photogenerated EHPs of hematite nanoflakes and that of hematite thin film. Taking advantage of this effect, the Sn-HB 600 shows the 0.71 mA cm$^{-2}$ at 1.23 $V_{RHE,}$ which is 10.52 times higher photocurrent density compared with HNF-600 (0.0675 mA cm$^{-2}$ at 1.23 $V_{RHE}$). Our study paves the way to reducing the recombination of thin film photoanode and also provides a straightforward way to solve the chronic issue of poor conductive sub-oxides bottom layer in thermal oxidized metals.

## Experimental Section

### Preparation of HNF and Sn-HB Samples

Iron thin films of 200, 400, 600, and 800 nm in thickness were prepared on FTO glass by using a magnetron sputtering. The iron films were deposited by an advanced plasma deposition method known as high-impulse magnetron sputtering (HiPIMS). The 4'' iron target (purity 99.99 %, Lesker) and Argon atmosphere were used as the source of iron and working gas, respectively. The films were deposited on carefully cleaned (rinsed in isopropyl alcohol, ethanol, and deionized water respectively) FTO coated glass substrates (Solaronix) at the room temperature (for the sake of clarity the temperature of the FTO substrate raised up to about 80 °C due to the plasma bombardment during the deposition). The FTO samples were placed in an ultra-high vacuum chamber on a rotating substrate holder. The depositions of iron were carried out with the applied Direct current (DC) power of 650 W using a pulsed mode with the pulse frequency of 100 Hz and duty cycle of 1 %. The iron film/FTO samples were placed in a furnace at 400 °C for 2 h in order to prepare the HNF samples. For Sn-doping, 20 μl of $SnCl_4$ ethanol solution (ethanol:$SnCl_4$ = 200 : 1) was dropped on the HNF samples and dried in air for 20 min, and then annealed at 800 °C in a furnace for 15 min for preparing Sn-HB samples.[36] To remove the excess $SnO_2$, the samples were immersed in 1 M KOH solution for 12 h. Additional samples were prepared on iron foil, without FTO substrate. The HNF on iron foil samples (i. e., hematite nanoflake on Fe without FTO) were prepared by the same procedure as the mentioned one above. The Sn-HB on iron foil samples were prepared by 5 min annealing time at 800 °C with $SnCl_4$ ethanol solution. In the text, the samples are coded according to the thickness of the sputtered iron films as follows: i) hematite nanoflakes grown during the first annealing step at 400 °C for two hours are named as HNF-#, # = 200, 400, 600, and 800; ii) fully oxidized and Sn-doped hematite nanoflakes achieved after the second annealing step at 800 °C for 15 min. are named as Sn-HB #, # = 200, 400, 600, and 800; and iii) for comparative reasons, the hematite nanoflakes grown from commonly used iron foils at 400 °C for two hours are named as HNF-iron foil.

### Preparation of Zn-Co LDHs

To synthesize Zn–Co LDHs, 10 ml deionized water (DI) solution, containing 44 mg of zinc nitrate hexahydrate (0.15 mmol), 87 mg of cobalt nitrate hexahydrate (0.3 mmol) and 144 mg of urea (2.4 mmol) were prepared and then 40 ml ethylene glycol was added here. This solution was placed in microwave reactor and treated microwave irradiation for 10 min with 30 s on/off interval and then cooling naturally. The Zn–Co LDH was filtered, washed with DI water, ethanol, and dried at 60 °C overnight.

### Preparation of Zn-Co LDH/Sn-HB 600

The Zn–Co LDH powder was dispersed in DI water with a density of 0.1 mg ml$^{-1}$. After Zn–Co LDH solution was sonicated for 10 min, Sn-HB 600 was immersed in Zn–Co LDH solution for 10 min. The Zn–Co LDH/Sn-HB 600 was washed by DI water and dried with $N_2$ gas.

### Photoelectrochemical measurements

The photoelectrochemical performance of the hematite photo-anodes was measured in a three-electrode PEC cell, where a Pt mesh and a Ag/AgCl (3 M KCl) electrode acted as a counter and reference electrode, respectively, under illumination of AM1.5G (100 mW cm$^{-2}$, 1 sun) in 1 M KOH electrolyte. Photocurrent density vs. applied potential (*J-V*) was measured by scanning the potential from −0.5 to 0.7 V at a scan rate of 2 mV s$^{-1}$. The potentials vs. Ag/AgCl (3 M KCl) were converted to the reversible hydrogen electrode (RHE) via the following equation $E_{RHE} = E_{Ag/AgCl} + 0.059pH + E^0_{Ag/AgCl}$, where $E_{Ag/AgCl}$ is the experimentally measured potential, and $E^0_{Ag/AgCl} = 0.209$ V at 25 °C for an Ag/AgCl electrode in 3 M KCl. IPCE was acquired in the range from 300 to 700 nm with 10 nm steps at an applied potential of 1.23 $V_{RHE}$ in 1 M KOH. The EIS measurements were carried out in the frequency range from 100 kHz to 0.1 Hz at 1.23 $V_{RHE}$ with a perturbation amplitude of 10 mV in a three-electrode set-up by using a 369 nm LED light source at 0.335 mW m$^{-2}$.

## Acknowledgements

*We would like to acknowledge the ERC, the DFG, the Erlangen DFG cluster of excellence EAM, project EXC 315 (Bridge), the DFG funCOS and the Operational Programme Research, Development and Education-European Regional Development Fund, project no. CZ.02.1.01/0.0/0.0/15_003/0000416. We acknowledge the funding from Czech Science Foundation, project GA CR – EXPRO, 19–27454X. Open Access funding enabled and organized by Projekt DEAL.*

## Conflict of Interest

The authors declare no conflict of interest.

## Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.


© 2022 The Authors. ChemElectroChem published by Wiley-VCH GmbH.

**Keywords:** Hematite nanoflake · $Fe_3O_4$ · PEC water splitting · Recombination · Sn doping

[1] S. Chu, A. Majumdar, *Nature* **2012**, *488*, 294.
[2] H. J. Ahn, K. Y. Yoon, M. J. Kwak, J. S. Lee, P. Thiyagarajan, J. H. Jang, *J. Mater. Chem. A* **2015**, *3*, 21444.
[3] I. S. Cho, H. S. Han, M. Logar, J. Park, X. Zheng, *Adv. Energy Mater.* **2016**, *6*, 1501840.
[4] N. T. Nguyen, M. Altomare, J. Yoo, P. Schmuki, *Adv. Mater.* **2015**, *27*, 3208.
[5] H.-J. Ahn, K.-Y. Yoon, M.-J. Kwak, J.-H. Jang, *Angew. Chem. Int. Ed.* **2016**, *55*, 9922.
[6] P. Thiyagarajan, H.-J. Ahn, J.-S. Lee, J.-C. Yoon, J.-H. Jang, *Small* **2013**, *9*, 2341.
[7] L. Wang, N. T. Nguyen, X. Zhou, I. Hwang, M. S. Killian, P. Schmuki, *ChemSusChem* **2015**, *8*, 2615.
[8] X. Liu, F. Wang, Q. Wang, *Phys. Chem. Chem. Phys.* **2012**, *14*, 7894.
[9] T. W. Kim, K.-S. Choi, *Science.* **2014**, *343*, 990.
[10] K. Sivula, F. Le Formal, M. Gratzel, *ChemSusChem* **2011**, *4*, 432.
[11] S. D. Tilley, M. Cornuz, K. Sivula, M. Grätzel, *Angew. Chem. Int. Ed.* **2010**, *49*, 6405.
[12] J. Y. Kim, G. Magesh, D. H. Youn, J.-W. Jang, J. Kubota, K. Domen, J. S. Lee, *Sci. Rep.* **2013**, *3*, 2681.
[13] J. H. Kennedy, K. W. Frese, *J. Electrochem. Soc.* **1978**, *125*, 709.
[14] H.-J. Ahn, A. Goswami, F. Riboni, S. Kment, A. Naldoni, S. Mohajernia, R. Zboril, P. Schmuki, *ChemSusChem* **2018**, *11*, 1873.
[15] F. Li, J. Jian, Y. Xu, S. Wang, H. Wang, H. Wang, *Eng. Rep.* **2021**, *3*, e12387.
[16] H.-J. Ahn, M.-J. Kwak, J.-S. Lee, K.-Y. Yoon, J.-H. Jang, *J. Mater. Chem. A* **2014**, *2*, 19999.
[17] O. Zandi, B. M. Klahr, T. W. Hamann, *Energy Environ. Sci.* **2013**, *6*, 634.
[18] M. G. Ahmed, I. E. Kretschmer, T. A. Kandiel, A. Y. Ahmed, F. A. Rashwan, D. W. Bahnemann, *ACS Appl. Mater. Interfaces* **2015**, *7*, 24053.
[19] S. Kment, F. Riboni, S. Pausova, L. Wang, L. Wang, H. Han, Z. Hubicka, J. Krysa, P. Schmuki, R. Zboril, *Chem. Soc. Rev.* **2017**, *46*, 3716.
[20] M. Li, Y. Yang, Y. Ling, W. Qiu, F. Wang, T. Liu, Y. Song, X. Liu, P. Fang, Y. Tong, Y. Li, *Nano Lett.* **2017**, *17*, 2490.
[21] K.-Y. Yoon, H.-J. Ahn, M.-J. Kwak, S.-I. Kim, J. Park, J.-H. Jang, *J. Mater. Chem. A* **2016**, *4*, 18730.
[22] K. Zhang, X. Shi, J. K. Kim, J. S. Lee, J. H. Park, *Nanoscale* **2013**, *5*, 1939.
[23] P. Kumar, P. Sharma, R. Shrivastav, S. Dass, V. R. Satsangi, *Int. J. Hydrogen Energy* **2011**, *36*, 2777.
[24] C. Miao, T. Shi, G. Xu, S. Ji, C. Ye, *ACS Appl. Mater. Interfaces* **2013**, *5*, 1310.
[25] A. Kay, I. Cesar, M. Grätzel, *J. Am. Chem. Soc.* **2006**, *128*, 15714.
[26] D. K. Zhong, D. R. Gamelin, *J. Am. Chem. Soc.* **2010**, *132*, 4202.
[27] R. Liu, Z. Zheng, J. Spurgeon, X. Yang, *Energy Environ. Sci.* **2014**, *7*, 2504.
[28] L. Wang, C.-Y. Lee, A. Mazare, K. Lee, J. Müller, E. Spiecker, P. Schmuki, *Chem. Eur. J.* **2013**, *20*, 77.
[29] P. M. Rao, X. Zheng, *Nano Lett.* **2009**, *9*, 3001.
[30] T. Vincent, M. Gross, H. Dotan, A. Rothschild, *Int. J. Hydrogen Energy* **2012**, *37*, 8102.
[31] B. D. Chernoordik, H. B. Russell, U. Cvelbar, J. B. Jasinski, V. Kumar, T. Deutsch, M. K. Sunkara, *Nanotechnology* **2012**, *23*, 194009.
[32] C. Zhu, C. Li, M. Zheng, J.-J. Delaunay, *ACS Appl. Mater. Interfaces* **2015**, *7*, 22355.
[33] L. Wang, N. T. Nguyen, Y. Zhang, Y. Bi, P. Schmuki, *ChemSusChem* **2017**, *10*, 2720.
[34] J. L. Meijering, *Acta Metall.* **1955**, *3*, 157.
[35] Y. Xie, Y. Ju, Y. Toku, Y. Morita, *R. Soc. Open Sci.* **2018**, *5*, 172126.
[36] L. Xi, S. Y. Chiam, W. F. Mak, P. D. Tran, J. Barber, S. C. J. Loo, L. H. Wong, *Chem. Sci.* **2013**, *4*, 164.
[37] R. H. Gonçalves, E. R. Leite, *Energy Environ. Sci.* **2014**, *7*, 2250.
[38] T. P. Almeida, M. W. Fay, T. W. Hansen, Y. Zhu, P. D. Brown, *CrystEngComm* **2014**, *16*, 1540.
[39] G. Ruan, S. Wu, P. Wang, J. Liu, Y. Cai, Z. Tian, Y. Ye, C. Liang, G. Shao, *RSC Adv.* **2014**, *4*, 63408.
[40] S. Song, J. Kim, D. Lee, J. Lee, T. Min, J.-A. Chae, J.-S. Bae, J. Lee, J.-S. Lee, S. Park, *J. Am. Ceram. Soc.* **2017**, *100*, 3928.
[41] J. Velev, A. Bandyopadhyay, W. H. Butler, S. Sarker, *Phys. Rev. B* **2005**, *71*, 205208.
[42] S. Kment, P. Schmuki, Z. Hubicka, L. Machala, R. Kirchgeorg, N. Liu, L. Wang, K. Lee, J. Olejnicek, M. Cada, I. Gregora, R. Zboril, *ACS Nano* **2015**, *9*, 7113.
[43] H. Dotan, K. Sivula, M. Grätzel, A. Rothschild, S. C. Warren, *Energy Environ. Sci.* **2011**, *4*, 958.

Manuscript received: January 19, 2022
Revised manuscript received: February 27, 2022
Accepted manuscript online: March 14, 2022


© 2022 The Authors. ChemElectroChem published by Wiley-VCH GmbH.